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## RESEARCH INTERESTS

Visual world models, self-supervised visual learning, 3D vision, vision-language models, and diagnostic evaluation. I study how visual systems can learn predictive representations from time, geometry, and action, and how to evaluate whether those representations support reasoning and interaction beyond benchmark shortcuts.

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## EDUCATION

4/2025 – 3/2027 (expected) **M.Eng. in Computer Science** University of Tsukuba, Japan  
Advisor: [Prof. Yutaka Satoh](#). CGPA: **4.27/4.30**.

4/2021 – 3/2025 **B.Eng. in Applied Mechanics and Aerospace Engineering** Waseda University, Japan  
Minor in Computer Science. WES-evaluated GPA: **3.29/4.00**. Undergraduate research on reinforcement learning for energy-system optimization.

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## PEER-REVIEWED PUBLICATIONS

1. **Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Xianzheng Ma, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
**Beyond Single Object: Learning 3D Relations with Large Language Models.**  
IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (**CVPR Findings**), 2026.  
[project](#) | [paper](#) | [code](#)
2. Ryosuke Yamada, **Kohsuke Ide**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Gilles Puy, Andrei Bursuc, and Yuki M. Asano.  
**3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds.**  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.  
[project](#) | [paper](#) | [code](#)
3. **Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
**Seeing Red, Thinking Bad: Color Bias in Vision Language Models.**  
International Conference on Pattern Recognition (**ICPR**), 2026.  
[project](#) | [paper](#) | [code](#)
4. **Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
**Colors You Can't See, Semantic Biases You Can't Ignore.**  
ICCV Workshop on Multimodal Reasoning and General Intelligence (**ICCV Workshop**), 2025.
5. **Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, and Yutaka Satoh.  
**Can 3D Large Language Models Count to Three?**  
Meeting on Image Recognition and Understanding (**MIRU**), 2025.  
**Interactive Presentation Award (Top 4%).**
6. **Kohsuke Ide**, Ryosuke Yamada, Itsuki Ueda, and Hirokatsu Kataoka.  
**Efficient 3D Language Fields with Vector Quantized Gaussian Splatting.**  
Symposium on Sensing via Image Information (**SSII**), 2024.

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## RESEARCH EXPERIENCE

9/2025 – Present **Research Assistant** AIST, Japan  
Research on visual world models, 3D understanding, vision-language models, and evaluation. Lead projects from problem formulation and implementation through experiments and writing. This work has produced first-author papers at CVPR Findings and ICPR and an international collaboration with Oxford's Visual Geometry Group.

1/2024 – 8/2025 **Technical Trainee** AIST, Japan  
Developed diagnostic evaluations and datasets for 3D and vision-language models, including tests for shortcut learning, language leakage, and visual biases. Built research prototypes and experimental pipelines for multi-object 3D reasoning and scalable 3D pre-training.

## RESEARCH EXPERIENCE (CONTINUED)

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- 8/2024 – 3/2025 **Research Intern** Preferred Networks, Japan  
Worked on 3D reconstruction and free-viewpoint video technologies.
- 9/2021 – 3/2024 **Machine Learning Engineer Intern** Lightblue Technology, Japan  
Developed object-tracking systems for edge devices and improved a human-action-recognition pipeline through feature and optimization design.
- 9/2023 – 10/2023 **MLOps Engineer Intern** M3, Japan  
Developed and open-sourced a system for automatically recovering Kubernetes jobs from out-of-memory failures.

## RESEARCH VISITS & INTERNATIONAL COLLABORATION

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- 1/19 – 1/31/2026 **Research Visit** University of Oxford, UK  
Research meetings with members of the Visual Geometry Group and Active Vision Laboratory, including Xianzheng Ma and Christian Rupprecht.
- 4/22 – 4/30/2026 **ASPIRE Computer Vision Workshop and Research Visit** University of Oxford, UK  
Presented research and discussed follow-up work with researchers at Oxford VGG.

## ACADEMIC SERVICE

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- **Workshop Organizer**, [Visual General Intelligence: Vision Research Toward the AGI Era](#), CVPR 2026.

## HONORS & AWARDS

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- 7/2025 **MIRU 2025 Interactive Presentation Award** Top 4%  
Awarded for "Can 3D Large Language Models Count to Three?"
- 2023 **WINC Hackathon Winner**  
Built an edge-device push-up counter using computer vision.

## TECHNICAL SKILLS & LANGUAGES

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- **Research:** Computer vision, visual world models, self-supervised learning, 3D reconstruction, vision-language models, diagnostic evaluation.
- **Programming and systems:** Python, C++, PyTorch, CUDA, OpenCV, Git, Docker, Kubernetes, Linux, AWS, and GCP.
- **Languages:** Japanese (native); English (fluent; TOEFL iBT 118/120, TOEIC 990/990).